

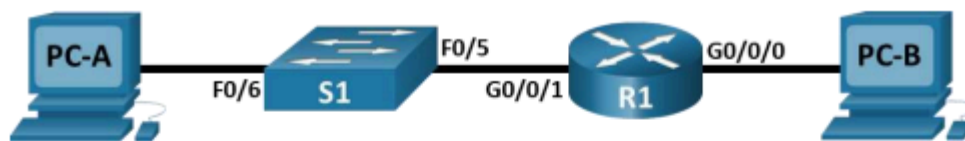
# Arbeitsbericht

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- 

## Topologie:

### Topologie



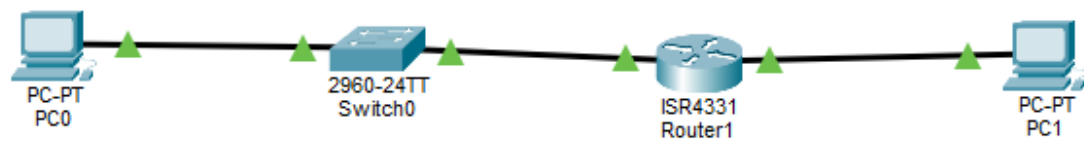
### Adresstabelle

| Device | Interface | IP Address / Prefix   | Default Gateway |
|--------|-----------|-----------------------|-----------------|
| R1     | G0/0/0    | 192.168.0.1 /24       | N/A             |
|        |           | 2001:db8:acad::1/64   |                 |
|        |           | fe80::1               |                 |
|        | G0/0/1    | 192.168.1.1 /24       | N/A             |
|        |           | 200:db8:acad:1::1/64  |                 |
|        |           | fe80::1               |                 |
| S1     | VLAN 1    | 192.168.1.2 /24       | 192.168.1.1     |
| PC-A   | NIC       | 192.168.1.3 /24       | 192.168.1.1     |
|        |           | 2001:db8:acad:1::3/64 | fe80::1         |
| PC-B   | NIC       | 192.168.0.3 /24       | 192.168.0.1     |
|        |           | 2001:db8:acad::3/64   | fe80::1         |

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## Ziele:

- Teil 1: Einrichten der Topologie und Initialisieren von Geräten



- Teil 2: Konfigurieren von Geräten und Überprüfen der Konnektivität

- Schritt 1** Konfigurationen der Geräte:

### PC-A

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 0.0.0.0

IPv6 Configuration

☒ Automatic ☐ Static

IPv6 Address 2001:DB8:ACAD:1:202:4AFF:FE6E:1B40 / 64

Link Local Address FE80::202:4AFF:FE6E:1B40

Default Gateway FE80::1

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MDS

Username

Password

### PC-B

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.0.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☒ Automatic ☐ Static

IPv6 Address 2001:DB8:ACAD:0:250:FFF:FEA1:659D / 64

Link Local Address FE80::250:FFF:FEA1:659D

Default Gateway FE80::1

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MDS

Username

Password

**S1**

**R1**

```
enable secret 5 $!$mERr$9cTjUIEqNGurQiFU.ZeCi1
enable password cisco
!
!
!
!
!
!
ip cef
ipv6 unicast-routing
!
ipv6 cef
!
!
!
interface GigabitEthernet0/2
!
interface Vlan1
!
ip address 192.168.1.2 255.255.255.0
!
ip default-gateway 192.168.1.1
!
!
!
!
!
!
line con 0
password cisco
!
line vty 0 4
login
line vty 5 15
login
!
!
!
!
end

spanning-tree mode pvst
!
!
!
!
interface GigabitEthernet0/0/0
ip address 192.168.0.1 255.255.255.0
duplex auto
speed auto
ipv6 address FE80::1 link-local
ipv6 address 2001:DB8:ACAD::1/64
!
interface GigabitEthernet0/0/1
ip address 192.168.1.1 255.255.255.0
duplex auto
speed auto
ipv6 address FE80::1 link-local
ipv6 address 200:DB8:ACAD:1::1/64
```

### Befehle für S1 Konfiguration

```
S1# conf t
S1(config)# int vlan1
S1(config-if)# ip addr 192.168.1.2 255.255.255.0
S1(config-if)# exit

S1(config)#ip default-gateway 192.168.1.1
S1(config)# sdm prefer dual-ipv4-and-ipv6 default
S1(config)# line console 0
S1(config-line)# password cisco
S1(config-line)# login
S1(config-line)# exit

S1(config)# enable secret class
S1(config)# end

S1# copy running-config startup-config
```

**Befehle für R1 Konfiguration:**

```

R1# conf t
R1(config)# int g0/0/0
R1(config-if)# ip addr 192.168.0.1 255.255.255.0
R1(config-if)# ipv6 addr 2001:db8:acad::1/64
R1(config-if)# ipv6 addr fe80::1 link-local
R1(config-if)# no shutdown
R1(config-if)# exit

R1(config)# int g0/0/1
R1(config-if)# ip addr 192.168.1.1 255.255.255.0
R1(config-if)# ipv6 addr 200:db8:acad:1::1/64
R1(config-if)# ipv6 addr fe80::1 link-local
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# ipv6 unicast-routing

R1(config)# line console 0
R1(config-line)# password cisco
R1(config-line)# login
R1(config-line)# exit
R1(config)# enable secret class
R1(config)# end

R1# copy running-config startup-config

```

- **Schritt 2** Überprüfen der Konnektivität:

**PC-A -> PC-B**

```

C:\>ping 192.168.0.3

Pinging 192.168.0.3 with 32 bytes of data:

Reply from 192.168.0.3: bytes=32 time<1ms TTL=127
Reply from 192.168.0.3: bytes=32 time=10ms TTL=127
Reply from 192.168.0.3: bytes=32 time<1ms TTL=127
Reply from 192.168.0.3: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

```

**S1 -> PC-B**

```

Switch#ping 192.168.0.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.0.3, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

```

```

C:\>ping 2001:DB8:ACAD:0:250:FFF:FEA1:659D

Pinging 2001:DB8:ACAD:0:250:FFF:FEA1:659D with 32 bytes of data:

Reply from 2001:DB8:ACAD:0:250:FFF:FEA1:659D: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ACAD:0:250:FFF:FEA1:659D: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ACAD:0:250:FFF:FEA1:659D: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ACAD:0:250:FFF:FEA1:659D: bytes=32 time<1ms TTL=127

Ping statistics for 2001:DB8:ACAD:0:250:FFF:FEA1:659D:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

```

- **Schritt 3** Geräteinformationen anzeigen:

Router:

**show Befehl****Bild**

show ip route

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```

      192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.0.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.0.1/32 is directly connected, GigabitEthernet0/0/0
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0/1
```

show ipv6  
route

```
Router#show ipv6 route
IPv6 Routing Table - 5 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
C 200:DB8:ACAD:1::/64 [0/0]
   via GigabitEthernet0/0/1, directly connected
L 200:DB8:ACAD:1::1/128 [0/0]
   via GigabitEthernet0/0/1, receive
C 2001:DB8:ACAD::/64 [0/0]
   via GigabitEthernet0/0/0, directly connected
L 2001:DB8:ACAD::1/128 [0/0]
   via GigabitEthernet0/0/0, receive
L FF00::/8 [0/0]
   via Null0, receive
```

Switch:

**show Befehl****Bild**

show ip interface

```

Switch#show ip interface
Vlan1 is up, line protocol is up
Internet address is 192.168.1.2/24
Broadcast address is 255.255.255.255
Address determined by setup command
MTU is 1500 bytes
Helper address is not set
Directed broadcast forwarding is disabled
Outgoing access list is not set
Inbound access list is not set
Proxy ARP is enabled
Local Proxy ARP is disabled
Security level is default
Split horizon is enabled
ICMP redirects are always sent
ICMP unreachable are always sent
ICMP mask replies are never sent
IP fast switching is disabled
IP fast switching on the same interface is disabled
IP Null turbo vector
IP multicast fast switching is disabled
IP multicast distributed fast switching is disabled
IP route-cache flags are None
Router Discovery is disabled
IP output packet accounting is disabled
IP access violation accounting is disabled
TCP/IP header compression is disabled
RTP/IP header compression is disabled
Probe proxy name replies are disabled
Policy routing is disabled
Network address translation is disable
WCCP Redirect outbound is disabled
WCCP Redirect inbound is disabled
WCCP Redirect exclude is disabled
BGP Policy Mapping is disabled

```

- **Schritt 4** Zeigen Sie die Routing-Tabelle auf dem Router an

a.

```

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.0.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.0.1/32 is directly connected, GigabitEthernet0/0/0
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0/1

```

Welcher Code wird in der Routing-Tabelle verwendet, um ein direkt verbundenes Netzwerk anzuzeigen?

C

Wie viele Route-Einträge sind in der Routing-Tabelle mit einem C-Code codiert?

## 2 Einträge

Welche Schnittstellentypen sind den C-codierten Routen zugeordnet?

## GigabitEthernet

b.

```
Router#show ipv6 route
IPv6 Routing Table - 5 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
        U - Per-user Static route, M - MIPv6
        I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
        ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
        O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
        ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
        D - EIGRP, EX - EIGRP external
C   200:DB8:ACAD:1::/64 [0/0]
    via GigabitEthernet0/0/1, directly connected
L   200:DB8:ACAD:1::1/128 [0/0]
    via GigabitEthernet0/0/1, receive
C   2001:DB8:ACAD::/64 [0/0]
    via GigabitEthernet0/0/0, directly connected
L   2001:DB8:ACAD::1/128 [0/0]
    via GigabitEthernet0/0/0, receive
L   FF00::/8 [0/0]
    via Null0, receive
```

Welcher Code wird in der Routing-Tabelle verwendet, um ein direkt verbundenes Netzwerk anzuzeigen?

C

Wie viele Route-Einträge sind in der Routing-Tabelle mit einem C-Code codiert?

## 2 Einträge

Welche Schnittstellentypen sind den C-codierten Routen zugeordnet?

## GigabitEthernet

- **Schritt 5** Schnittstelleninformationen auf dem Router R1 anzeigen.

a. `show ip int g0/0/1`

```
Router#show ip interface g0/0/1
GigabitEthernet0/0/1 is up, line protocol is up (connected)
  Internet address is 192.168.1.1/24
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is not set
  Directed broadcast forwarding is disabled
  Outgoing access list is not set
  Inbound access list is not set
  Proxy ARP is enabled
  Security level is default
  Split horizon is enabled
  ICMP redirects are always sent
  ICMP unreachable are always sent
  ICMP mask replies are never sent
  IP fast switching is disabled
  IP fast switching on the same interface is disabled
  IP Flow switching is disabled
  IP Fast switching turbo vector
  IP multicast fast switching is disabled
  IP multicast distributed fast switching is disabled
  Router Discovery is disabled
  IP output packet accounting is disabled
  IP access violation accounting is disabled
  TCP/IP header compression is disabled
  RTP/IP header compression is disabled
  Probe proxy name replies are disabled
  Policy routing is disabled
  Network address translation is disabled
  BGP Policy Mapping is disabled
  Input features: MCI Check
  WCCP Redirect outbound is disabled
  WCCP Redirect inbound is disabled
  WCCP Redirect exclude is disabled
```

Wie ist der Betriebszustand der G0/0/1-Schnittstelle?

**up**

Wie lautet die MAC-Adresse (Media Access Control) der G0/1-Schnittstelle?

**00d0.baa1.ec02 (wurde mit dem Befehl "show interfaces" gefunden)**

Wie wird die Internetadresse in diesem Befehl angezeigt?

**192.168.1.1/24**

b. `show ipv6 int g0/0/1`

```
Router#show ipv6 int g0/0/1
GigabitEthernet0/0/1 is up, line protocol is up
IPv6 is enabled, link-local address is FE80::1
No Virtual link-local address(es):
Global unicast address(es):
  200:DB8:ACAD:1::1, subnet is 200:DB8:ACAD:1::/64
Joined group address(es):
  FF02::1
  FF02::2
  FF02::1:FF00:1
MTU is 1500 bytes
ICMP error messages limited to one every 100 milliseconds
ICMP redirects are enabled
ICMP unreachable are sent
ND DAD is enabled, number of DAD attempts: 1
ND reachable time is 30000 milliseconds
ND advertised reachable time is 0 (unspecified)
ND advertised retransmit interval is 0 (unspecified)
ND router advertisements are sent every 200 seconds
ND router advertisements live for 1800 seconds
ND advertised default router preference is Medium
Hosts use stateless autoconfig for addresses.
Router#
```

Wie ist der Betriebszustand der G0/0/1-Schnittstelle?

**up**

Wie lautet die MAC-Adresse (Media Access Control) der G0/1-Schnittstelle?

**00d0.baa1.ec02 (wurde mit dem Befehl "show interfaces" gefunden)**

Wie wird die Internetadresse in diesem Befehl angezeigt?

**200:DB8:ACAD:1::1**

- **Schritt 6** Zeigen Sie eine zusammenfassende Liste der Schnittstellen auf dem Router und Switch an

`show ip interface brief`

```
Router#show ip int b
Interface          IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0/0 192.168.0.1    YES manual up          up
GigabitEthernet0/0/1 192.168.1.1    YES manual up          up
GigabitEthernet0/0/2 unassigned      YES unset   administratively down down
Vlan1              unassigned      YES unset   administratively down down
Router#
```

`show ipv6 interface brief`

```
Router#show ipv6 int b
GigabitEthernet0/0/0      [up/up]
    FE80::1
    2001:DB8:ACAD::1
GigabitEthernet0/0/1      [up/up]
    FE80::1
    200:DB8:ACAD:1::1
GigabitEthernet0/0/2      [administratively down/down]
    unassigned
Vlan1                     [administratively down/down]
    unassigned
Router#
```

show ip int

```
Switch#show ip int
Vlan1 is up, line protocol is up
  Internet address is 192.168.1.2/24
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is not set
  Directed broadcast forwarding is disabled
  Outgoing access list is not set
  Inbound access list is not set
  Proxy ARP is enabled
  Local Proxy ARP is disabled
  Security level is default
  Split horizon is enabled
  ICMP redirects are always sent
  ICMP unreachable are always sent
  ICMP mask replies are never sent
  IP fast switching is disabled
  IP fast switching on the same interface is disabled
  IP Null turbo vector
  IP multicast fast switching is disabled
  IP multicast distributed fast switching is disabled
  IP route-cache flags are None
  Router Discovery is disabled
  IP output packet accounting is disabled
  IP access violation accounting is disabled
  TCP/IP header compression is disabled
  RTP/IP header compression is disabled
  Probe proxy name replies are disabled
  Policy routing is disabled
  Network address translation is disable
  WCCP Redirect outbound is disabled
  WCCP Redirect inbound is disabled
  WCCP Redirect exclude is disabled
  BGP Policy Mapping is disabled
```

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## Reflexionsfragen

1. Wenn die G0/0/1-Schnittstelle anzeigt, dass sie administrativ ausgefallen ist, welchen Schnittstellenkonfigurationsbefehl würden Sie verwenden, um die Schnittstelle wieder hochzufahren?  
`no shutdown` in der Konfiguration der Interface G0/0/1
  2. Was würde passieren, wenn Sie die Schnittstelle G0/0/1 auf dem Router mit einer IP Adresse von 192.168.1.2 falsch konfiguriert hätten? PC-B wäre über den Router nicht erreichbar und könnte selbst nicht auf andere Hosts zugreifen.
-